

Nisarg Shah

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(Open to Relocate)

Results-driven **Data Analyst** with **4+** years of hands-on experience across healthcare, finance, and retail domains. Actively seeking data-driven roles where I can leverage analytics to solve real-world challenges and optimize outcomes.

Certifications & Recognitions

- **Machine Learning**
- Healthcare Assistant System Using Natural Language Understanding and Machine Learning - ISBN: 978-93-5391-770-8
- UAV Drone for Surveillance using Thermal Imaging - ISBN: 978-93-5437-776-1
- Stood 13th all over the world in the PDR (Preliminary design review) round, secured 2nd rank in the CDR (Critical design review) round, and bagged overall 9th position at the **CanSat Competition** organized by the American Astronautical Society and American Institute of Aeronautics and Astronautics in 2020
- Ranked 2nd in PDR (Preliminary design review) round and 2nd in CDR (Critical design review), securing overall 12th position in 2021

Experience

Rutgers New Jersey Medical School

Data Analyst

Remote, US

Jan 2024 - Present

Collaborated with biomedical researchers to accelerate drug discovery by engineering molecular features from chemical structures and deploying ML models that improved the prediction accuracy of molecular properties.

- Partnered with cross-functional **stakeholders**, including biomedical researchers and regulatory leads, to align data insights with business needs, inform R&D strategy, and support FDA-compliant reporting.
- Used **Excel** for exploratory analysis, molecular property audits, and validating feature extraction outputs to ensure data accuracy and regulatory compliance
- Automated molecular feature extraction and model inference workflows using **AWS Lambda**, enabling real-time scoring of incoming chemical data and streamlining the end-to-end prediction pipeline
- Developed **Power BI** dashboards to monitor compound performance metrics, model accuracy, and key chemical trends, supporting data-driven decisions across R&D and regulatory teams
- Optimized **SQL** queries and rebuilt ETL pipelines using **Python**, reducing data processing time and ensuring reliable, analysis-ready molecular property datasets.
- Built interactive dashboards using **Tableau** to visualize compound activity, model performance, and key molecular trends for stakeholders in research and compliance teams.
- Designed and implemented a full molecular feature extraction pipeline using **RDKit** and **DeepChem**, converting raw SMILES strings into rich chemical descriptors and fingerprint vectors for downstream modeling.
- Built and optimized classification models (**XGBoost**, **CatBoost**) for predicting drug efficacy, using **Optuna** for hyperparameter tuning, resulting in a **15%** improvement in classification accuracy.
- Deployed a **message-passing neural network (MPNN)** via **Chemprop** to model molecular structure-property relationships, enabling more accurate property predictions like molecule half-life.
- Conducted comparative model evaluation using statistical testing and cross-validation techniques to select the best-performing architectures.
- Computed **Morgan fingerprints (ECFP)** and RDKit descriptors such as TPSA, MolLogP, and NumHAcceptors to represent molecular structures numerically for machine learning.
- Designed quality control routines and feature validation steps to ensure data consistency and reliability in line with FDA and ICH data quality expectations.

Rutgers University

Data Analyst Teaching Assistant

New Jersey, US

May 2023 - Aug 2023

Rutgers aimed to identify at-risk students and enhance academic support by analyzing factors that influence student performance and outcomes.

- Used Python (pandas, NumPy, scikit-learn) to analyze student performance, applying regression analysis and clustering to detect patterns in GPA trends, course completion, and engagement.
- Conducted ANOVA and correlation analysis to assess how demographics and prior academic performance impacted success metrics, integrating insights into dashboards for academic advisors
- Extracted and transformed data using SQL to analyze historical and real-time enrollment trends, focusing on student demographics, program-specific enrollment, and retention rates.
- Taking recitations for the course: Computer Application for Business, graded assessment, and assisted in office hours

Simbott **Mumbai, India**
Data Analyst **Jul 2021 - Jul 2022**

Simbott uses AR/VR simulations to cut training time and costs. I improved device performance by training models with real-world usage data, analyzing trainee performance, conducting A/B tests, and building client segmentation models, which led to better training outcomes and a \$ 100K revenue boost through targeted outreach.

- Executed **A/B testing** to evaluate variations in simulator feedback mechanisms (e.g., visual cues, real-time corrections), leading to an **18%** increase in trainee skill acquisition rates.
- Applied **ANOVA** and **post-hoc statistical testing** on user satisfaction scores and performance metrics to identify significant UI/UX factors that improved engagement and reduced training dropout rates
- Developed real-time KPI dashboards in **Tableau** to visualize metrics like accuracy, module completion time, and error rates, helping instructors adapt training sessions in real-time
- Designed and implemented **SQL-based data pipelines** to store segmentation outputs and client metadata, enabling easy retrieval and dynamic filtering.
- Applied **K-Means clustering** on company data and training demand indicators to segment potential B2B clients, enabling targeted outreach and contributing to successful sales conversions for custom simulator deployments.

Pixeldust **Mumbai, India**
Data Analyst **Apr 2021 - Jun 2021**

Collaborated with a large retail clothing chain (10 locations) facing declining sales and competition from quick-commerce and online retailers; conducted customer feedback analysis and exploratory data analysis (EDA) on sales data to implement targeted solutions, resulting in an 18% month-over-month sales increase.

- Applied **natural language processing (NLP)** techniques using **SpaCy** to analyze thousands of customer reviews, uncovering key satisfaction drivers related to product quality, pricing, and delivery, leading to a **20%** improvement in customer retention
- Built a **text preprocessing** pipeline (tokenization, stopword removal, lemmatization) to clean and structure unstructured feedback data, enabling scalable NLP modeling.
- Designed an end-to-end **ETL pipeline** using **Apache Airflow** to ingest, clean, and transform large-scale retail transaction data into a **Snowflake** warehouse, reducing ingestion time by **30%**
- Developed a **Random Forest classifier** to identify high-value customers based on purchase patterns, recency-frequency-monetary (RFM) features, and engagement signals, enabling targeted marketing that increased revenue by **18%**.

Finbits **Mumbai, India**
Software Engineer **Jun 2020 - Mar 2021**

Finbits is a fintech media company where I built the platform from scratch to host recorded podcasts and developed a full email marketing funnel by analyzing user behavior and addressing drop-off points with data-driven UX improvements.

- Built a user registration and notification system using **Kafka** for event streaming, **Amazon SNS** for transactional email alerts, and **ElastiCache** to enable low-latency access to frequently requested data
- Developed and optimized **React/Redux**-based frontend for a fintech podcast and information platform, ensuring responsive design and cross-browser compatibility, which boosted customer engagement by **22%**
- Refactored **SQL** queries for high-traffic API endpoints to reduce database latency by **17%**, improving page load performance and user experience.
- Implemented user-centric features such as auto-login, real-time chat pop-ups, and a dynamic navbar to enhance usability and improve user retention metrics

Projects

Personalized AI Chatbot for Document Retrieval

- Integrated MiniLM V2 embedding model to generate document embeddings, improving the relevance of the response by 30%
- Developed interactive UI using OpenWebUI and integrated Milvus, enabling fast, accurate document retrieval
- Deployed a RAG Model, boosting the chatbot's accuracy in retrieving answers from PDFs

Google Stock Price Prediction

- Established a Recurrent Neural Network (RNN) model with LSTM layers using Keras to predict the opening price of Google stocks based on five years of data containing opening, high, low, and closing prices, achieving 85% accuracy
- Applied Min-Max Normalization and structured a 60-timestep sequence to capture temporal dependencies and enhance model accuracy.

Home Credit Default Risk

- Developed a custom Gini stability metric to evaluate model consistency, achieving a stability score of 0.472 by penalizing performance variability
- Addressed categorical inconsistencies through targeted encoding and imputation techniques, reducing test set error and enhancing model reliability
- Trained a stability-focused LightGBM model, achieving an ROC AUC of 0.751 on the validation set for risk prediction

Skills

- **Databases:** Snowflake, Microsoft SQL Server, MySQL, Amazon Redshift
- **Programming Languages:** Python (NumPy, Pandas, Matplotlib, Scikit-learn, PyTorch), R (Shiny, dplyr, ggplot2), JavaScript, HTML, CSS
- **Tools & Libraries:** AWS (Lambda, EC2, Glue, Sagemaker), Excel, PowerPoint, JIRA
- **Data Visualization:** PowerBI, Tableau, Tableau Prep Builder, Alteryx
- **Statistical Skills:** Descriptive Statistics, Hypothesis Tests, A/B Tests, ANOVA, Chi-Square Test
- **Machine Learning:** Supervised ML (Linear and Logistic Regression, Decision Trees, RF, Xgboost), Unsupervised ML (K-means, PCA), NLP, Classification techniques (binary classification, multi-class classification), A/B testing

Education

Rutgers University - New Brunswick

MS in Computer Science (GPA: 3.79/4)

New Jersey, US
Sep 2022 - May 2024

Dwarkadas J. Sanghvi College of Engineering

Bachelor of Engineering in Electronics and Telecommunication

Mumbai, India
Aug 2018 - Jun 2022